Frozen official DETR | Fixed exposure operation | Trainable FOCER module

(a) Original paper: fixed-exposure late fusion



HDR image

15 fixed exposures
stops 0, 1, …, 14



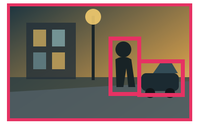
tone-mapped views

Detector
× 15

shared weights

15 detection sets

NMS
late box fusion



final detections

One complete detection pass per exposure
Information is combined only after boxes are predicted.

(b) FOCER-v2 inference: object-conditioned exposure routing

Each object query chooses its own useful exposure evidence.



HDR image

Fixed radiance scaling + gamma
15 views
stops 0, 1, …, 14
not learned; same as baseline



Shared ResNet-50 + DETR encoder
frozen
weights shared across views

15 encoded memories
spatial feature tokens remain separate by stop

FOCER routed DETR decoder layer × 6

100 object queries
current states

Query self-attention
queries communicate

Per-exposure cross-attention
15 evidence vectors/query

Exposure router
1. current query state
2. exposure evidence
3. stop embedding
4. local reliability

trainable: 1.25M parameters

per-query Top-3
different rows = different objects

Weighted evidence update
+
refined query

Frozen class + box heads
one coherent prediction set

No cross-exposure NMS
evidence fused inside queries

(c) Training-only guidance for learning meaningful routes

TRAINING ONLY - the teacher and labels are removed at inference

Ground-truth boxes + labels
training split only

Frozen DETR teacher
scores each labeled object under all 15 fixed exposures
guidance only; never deployed

Per-object exposure utility
Which stops help this object?
converted to a soft target

Routing supervision
teacher loss aligns each query's exposure preference with useful detector evidence

teaches the router

Trainable FOCER router
inside all 6 decoder layers

DETR detection losses
class + box + GIoU

What is optimized?
Only the added routing modules and their query updates
official DETR remains frozen

The teacher supplies exposure-selection knowledge; ordinary detection losses preserve the end task.

Core difference: the baseline fuses predicted boxes; FOCER-v2 lets each object query route and fuse exposure evidence before making one prediction.